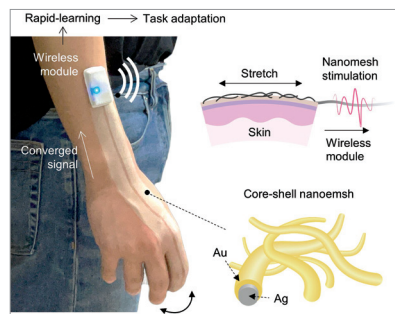




technologies provide the needed building blocks toward a more sustainable way to power the billions of sensor nodes in the near future. The team is currently working on the monolithic integration of these low-power devices with antenna and sensors to showcase their true potential.

Spray-on sensory system makes hands smarter

Researchers from Stanford University have developed a spray-on sensory system which consists of printed, bio-compatible nanomesh directly connected with a wireless



Spray-on sensory system which consists of printed, bio-compatible nanomesh directly connected with wireless Bluetooth module and further trained through meta-learning (Credit: Kyun Kyu Richard Kim, Bao Group, Stanford U.)

Bluetooth module and further trained through meta-learning. The device can identify objects by touch alone or allow users to communicate by hand gestures with apps in immersive environments. The spray-on approach enables the device to function without any need of

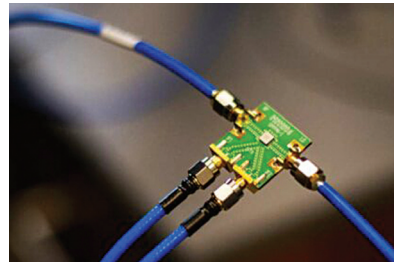
a substrate. This decision eliminated unwanted motion artifacts and allowed them to use a single trace of conductive mesh to generate multi-joint information of the fingers. The device functions on the basis of a machine learning algorithm. Computers monitor the changing patterns in conductivity and map those changes to specific physical tasks and gestures. To meet this intense computational challenge and the need to compute vast amounts of data, the Stanford team has developed a learning scheme that is far more computationally efficient.

Earbuds shown to work as hearing aids much more economically

Recent studies have demonstrated the effectiveness of earbuds in case of hearing loss. According to these studies, commercial earbuds can be adapted as hearing aids in amplifying sound for individuals with hearing loss. One such study was conducted by a team of researchers led by Yen-fu Cheng, an otolaryngologist at Taipei Veterans General Hospital. The team compared AirPods 2 and AirPods Pro—the model with a noise canceling feature—with a set of premium hearing aids and a basic pair of hearing aids. Notably, AirPods Pro met four out of five technology standards for hearing aids. It was also observed that both AirPods models were far cheaper than the basic as well as the premium hearing aids.

Wireless communication at terahertz speed shown over large distances

Researchers have proved that high-speed, high-bandwidth wireless communication at the terahertz frequency is possible across long distances. Their work shows that there is a path



Josep Jornet, an associate professor of electrical and computer engineering at Northeastern, proved it is possible to maintain a two kilometre-long connection using high frequency terahertz technology (Credit: Matthew Modoono, Northeastern University)

forward for mass wireless communication, one that could shrink the digital divide felt by rural communities outside high-speed optical fibre networks. Researchers were able to form a 2km link, the longest terahertz connection ever established on Earth.

The team directly fed the information to the signal source. The results showed that the new system was hitting frequencies and bandwidths that eclipsed 5G networks by more than “two orders of magnitude.” The impact of this kind of high-speed, high bandwidth connection would be monumental, providing higher data rates and more connectivity even for rural communities.

Solar reactor converts plastic and greenhouse gasses to fuel

Researchers at the University of Cambridge have developed a system that can transform plastic waste and greenhouse gasses into sustainable fuels and other valuable products—



Researchers have developed a system that can transform plastic waste and greenhouse gases into sustainable fuels and other valuable products—using just the energy from the Sun (Credit: University of Cambridge)

using just the energy from the Sun. It can convert two waste streams into two chemical products at the same time—the first time this has been achieved in a solar-powered reactor. The reactor converts the carbon dioxide and plastics into different products which can be

utilised in a variety of industries. Researchers were able to convert carbon dioxide into syngas, a key building block for sustainable liquid fuels, and plastic bottles were converted into glycolic acid, which is widely used in the cosmetics industry. They also designed different catalysts, which were integrated into the light absorber. Changing the catalyst enabled researchers to then change the end product.